

The business cycle and aggregate inventory investment

Part 1 · Why is it important? — solution report with data, method, results and charts

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1. The business cycle and aggregate inventory investment

Parameters: cycle length **24 months**, amplitude **±15 %**, production leads sales by **3 months**, starting inventory **150 units**, contribution margin **25 %**, holding cost **2 % per month**.

$$I(t) = I_0 + \sum [P(k) - S(k)]$$

183

Max inventory

95

Min inventory

88 units = 88% of a month's sales

Swing (max – min)

11.4

Largest monthly build-up

-11.4

Largest monthly disinvestment

22

Inventory peaks ≈ months after sales peak

Production leads sales → inventory builds before the peak and is drawn down after it

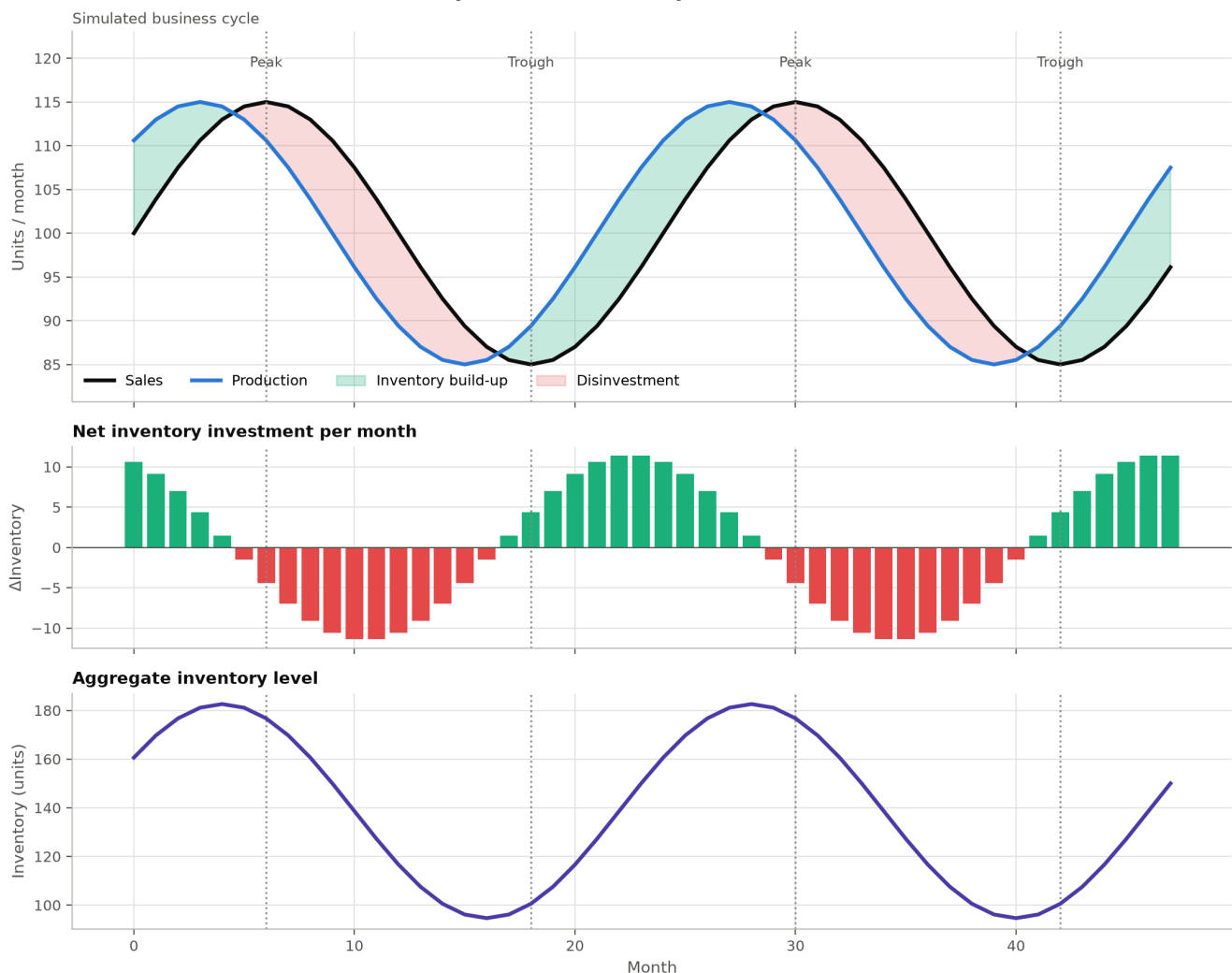


Figure 1.1 — Business-cycle simulation: sales vs production, monthly inventory investment and inventory level.

Corporate profits

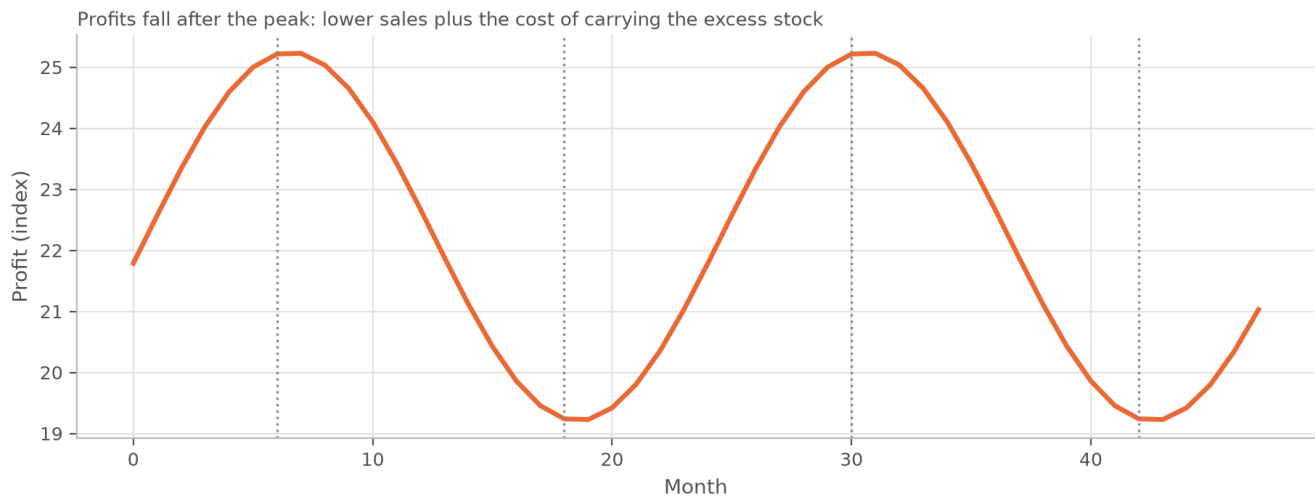


Figure 1.2 — Profit = margin × sales – holding cost × inventory.

Simulation output (every 3rd month).

month	sales	production	inventory change	inventory	profit
1	100.0	110.6	+10.6	160.6	21.79
4	110.6	115.0	+4.4	181.1	24.03
7	115.0	110.6	-4.4	176.7	25.22
10	110.6	100.0	-10.6	150.0	24.65
13	100.0	89.4	-10.6	116.6	22.67
16	89.4	85.0	-4.4	96.1	20.43
19	85.0	89.4	+4.4	100.5	19.24
22	89.4	100.0	+10.6	127.2	19.80
25	100.0	110.6	+10.6	160.6	21.79
28	110.6	115.0	+4.4	181.1	24.03
31	115.0	110.6	-4.4	176.7	25.22
34	110.6	100.0	-10.6	150.0	24.65
37	100.0	89.4	-10.6	116.6	22.67
40	89.4	85.0	-4.4	96.1	20.43
43	85.0	89.4	+4.4	100.5	19.24
46	89.4	100.0	+10.6	127.2	19.80